

L5 - 2.6 Inverse of a Function – Lesson

MCR3U

Inverse of a function:

- The inverse of a function f is denoted as f^{-1}
- The function and its inverse have the property that if $f(a) = b$, then $f^{-1}(b) = a$
- So if $f(5) = 13$, then $f^{-1}(13) = 5$
- More simply put: The inverse of a function has all the same points as the original function, except that the x 's and y 's have been reversed.

It is important to note that $f^{-1}(x)$ is read as "the inverse of f at x ". The -1 does not behave like an exponent.

$$f^{-1}(x) \neq \frac{1}{f(x)}$$



To draw an inverse, all you need to do is swap the x and y coordinates of each point.



Finding Inverses Numerically

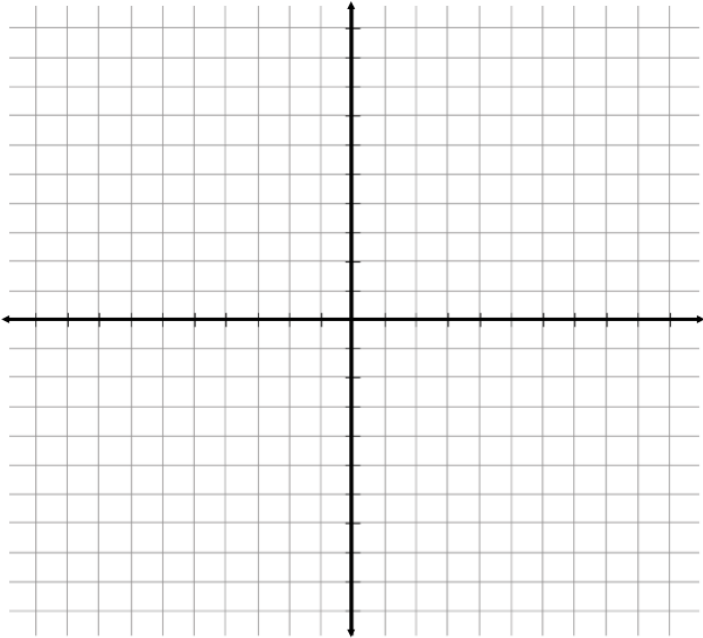
Example 1: The table shows ordered pairs belonging to a function $f(x)$. Determine $f^{-1}(x)$, then state the domain and range of $f(x)$ and its inverse.

$f(x)$	$f^{-1}(x)$
$(-5, 0)$	
$(-4, 2)$	
$(-3, 5)$	
$(-2, 6)$	
$(0, 7)$	

Example 2:

a) Graph the function $f(x) = x^2$ and its inverse $f^{-1}(x)$.

$f(x)$	$f^{-1}(x)$

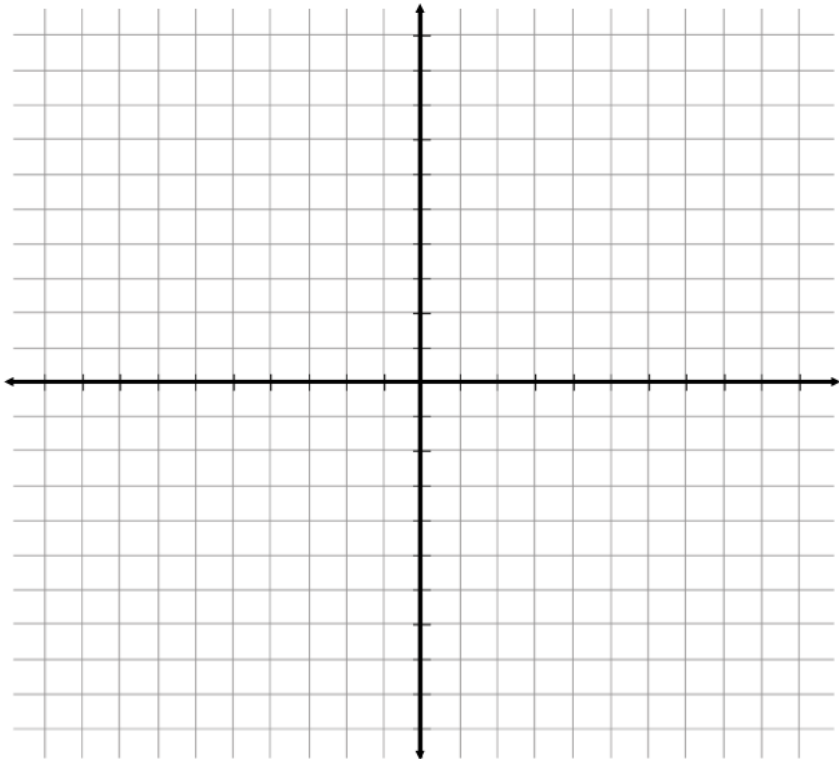


b) State the domain and range of both functions

Example 3: Sketch the graph of $g(x) = -2\sqrt{\left(-\frac{1}{2}x\right)} + 3$, then graph $g^{-1}(x)$.

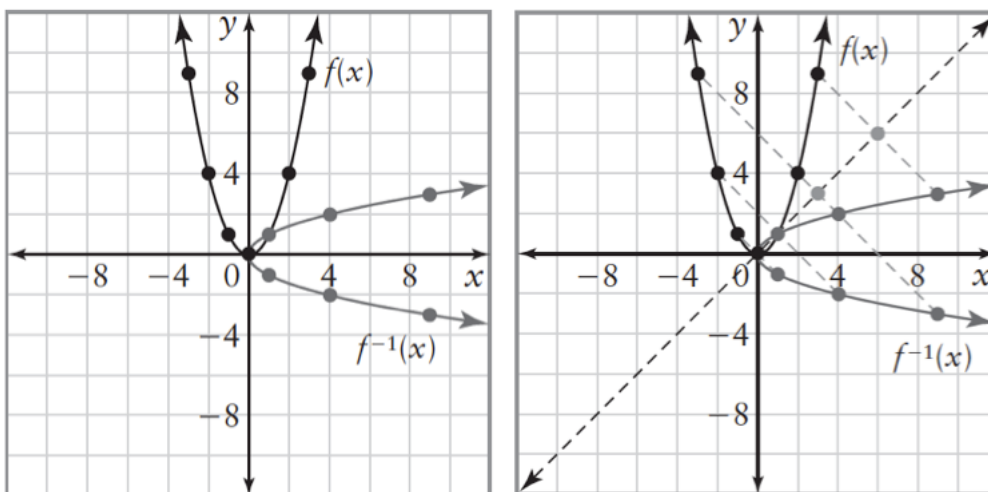
<i>x</i>	<i>y</i>

<i>x</i>	<i>y</i>

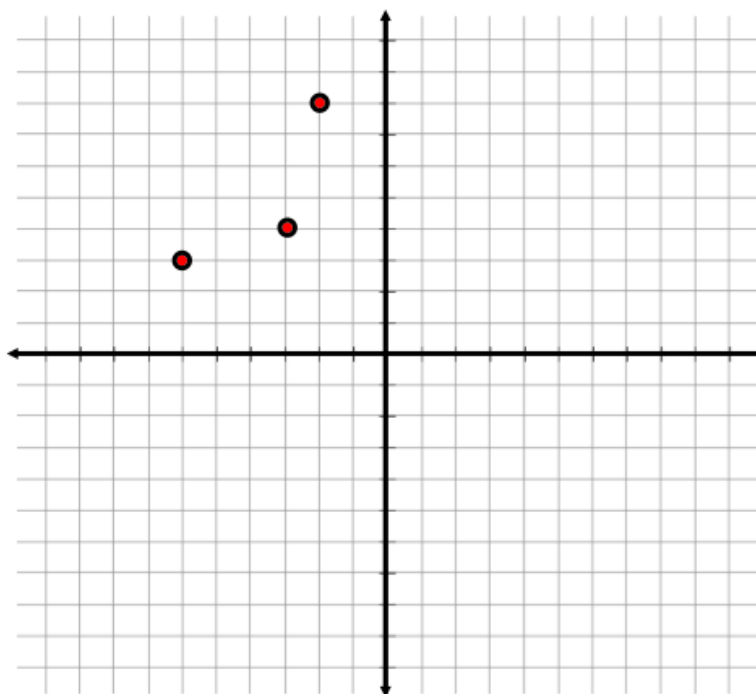


Finding Inverses by Graphing

The graph of $f^{-1}(x)$ is the graph of $f(x)$ reflected in the line $y = x$. This is true for all functions and their inverses. If you find the midpoint of each pair of points from example 2 and connect them you can prove this theorem.



Example 4: Sketch the inverse of $f(x)$



Finding Inverses Algebraically

Algebraic Method for finding the inverse:

1. Replace $f(x)$ with "y"
2. Switch the x and y variables
3. Isolate for y
4. replace y with $f^{-1}(x)$

a) $g(x) = \frac{3x}{4}$

b) $h(x) = 4x + 3$

c) $f(x) = x^2 - 1$

d) $h(x) = \frac{4x+3}{5}$

e) $f(x) = 2x^2 + 16x + 29$

Note: for algebraic inverses of quadratic functions, before interchanging x and y 's you must re-write in vertex form.

f) $r(x) = \sqrt{x} + 2$